Diary Data Analysis

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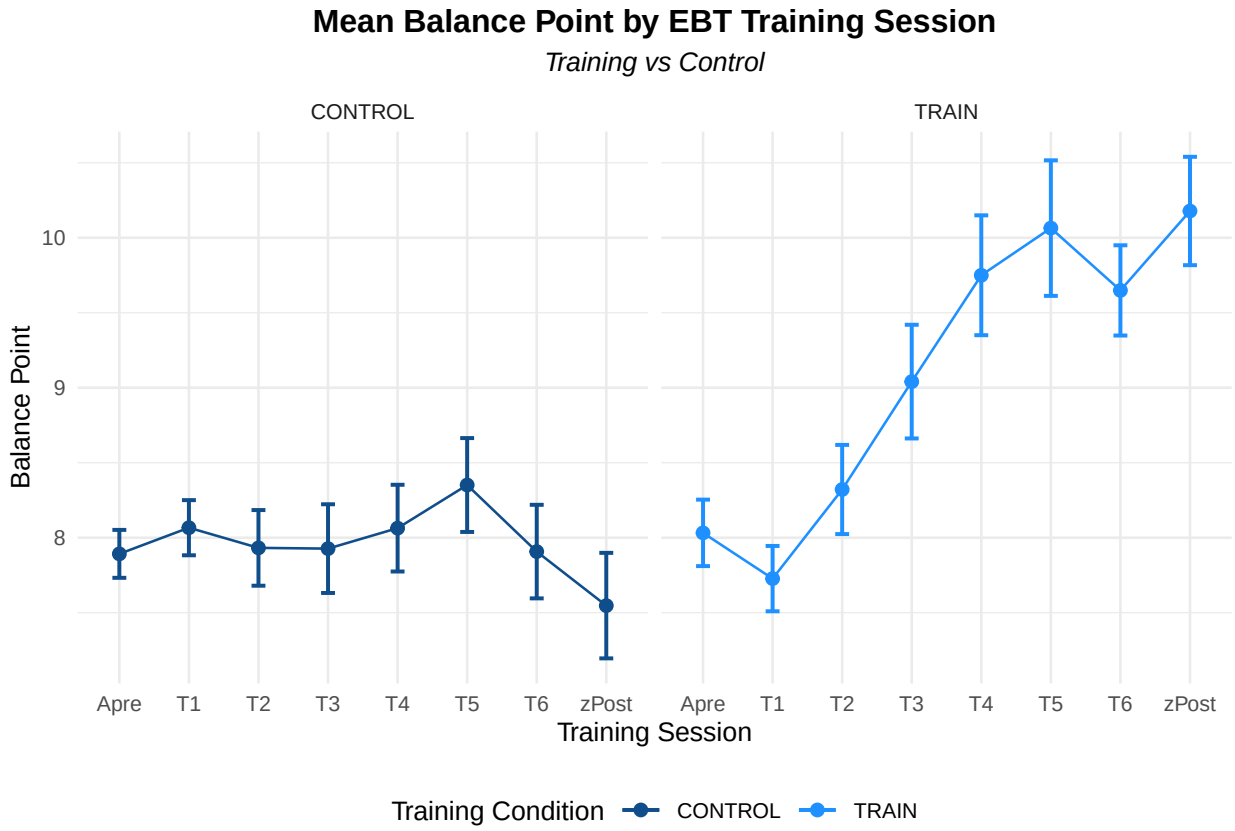
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1) Balance Point

1.1) Plotting data: Impact of training on balance point pre- and post-training

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by Training_Condition and aSession.
## i Output is grouped by Training_Condition.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(Training_Condition, aSession))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```



1.2) 1) Table of mean bp prepost by group

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by aSession and Training_Condition.
## i Output is grouped by aSession.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(aSession, Training_Condition))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```

aSession	Training_Condition	Group_N	Mean_bp	se_bp
Apre	CONTROL	51	7.823529	0.1649858
Apre	TRAIN	52	7.961538	0.2409157
zPost	CONTROL	51	7.607843	0.3587324
zPost	TRAIN	52	9.980769	0.3648285

1.2) Model 1 - Do children differ in their mean balancepoint at baseline?

1.2.1) Linear Regression Model

```
model_1_baseline <- lm(bp ~ Training_Condition, data = nep_baseline_only)
```

1.2.2) Results

```
##
## Call:
## lm(formula = bp ~ Training_Condition, data = nep_baseline_only)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.9615 -0.9615  0.0385  1.0385  3.1765
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                7.8235     0.2082  37.573   <2e-16 ***
## Training_ConditionTRAIN    0.1380     0.2931   0.471    0.639
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.487 on 101 degrees of freedom
## Multiple R-squared:  0.002191,   Adjusted R-squared:  -0.007688
## F-statistic: 0.2218 on 1 and 101 DF,  p-value: 0.6387
```

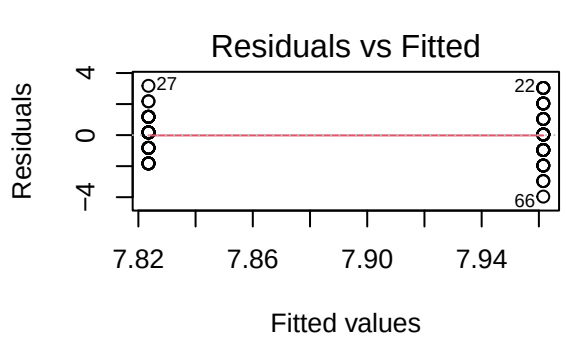
1.2.3) 95% Confidence Intervals

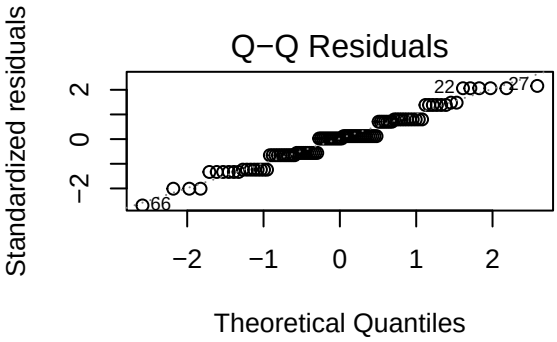
```
##                               2.5 %    97.5 %
## (Intercept)              7.4104674 8.2365914
## Training_ConditionTRAIN -0.4433336 0.7193517
```

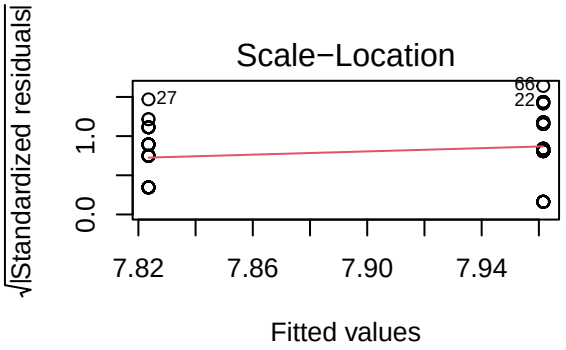
```
##                              0.25 %  99.75 %
## (Intercept)              7.2259224 8.421136
## Training_ConditionTRAIN -0.7030619 0.979080
```

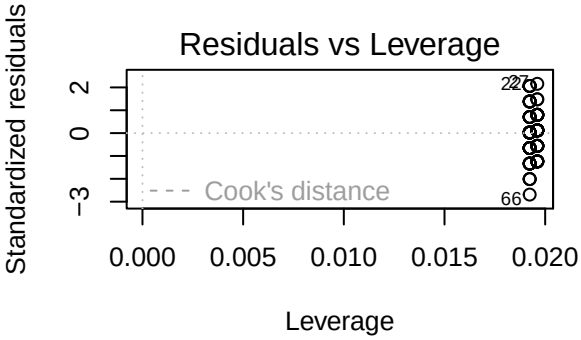
1.2.4) Assumption Checks:

Difference model









1.3) Model 2 - Do children differ in their mean balancepoint at at follow up?

Preregistered: Model predicting difference in T2 and T1 BP by IBT

1.3.1) Linear Regression Model

```
lm_bp_diff <- lm(difference_s2_minus_s1 ~ Training_Condition, data = nep_wide_pre_post_wider)
```

1.3.2) Results

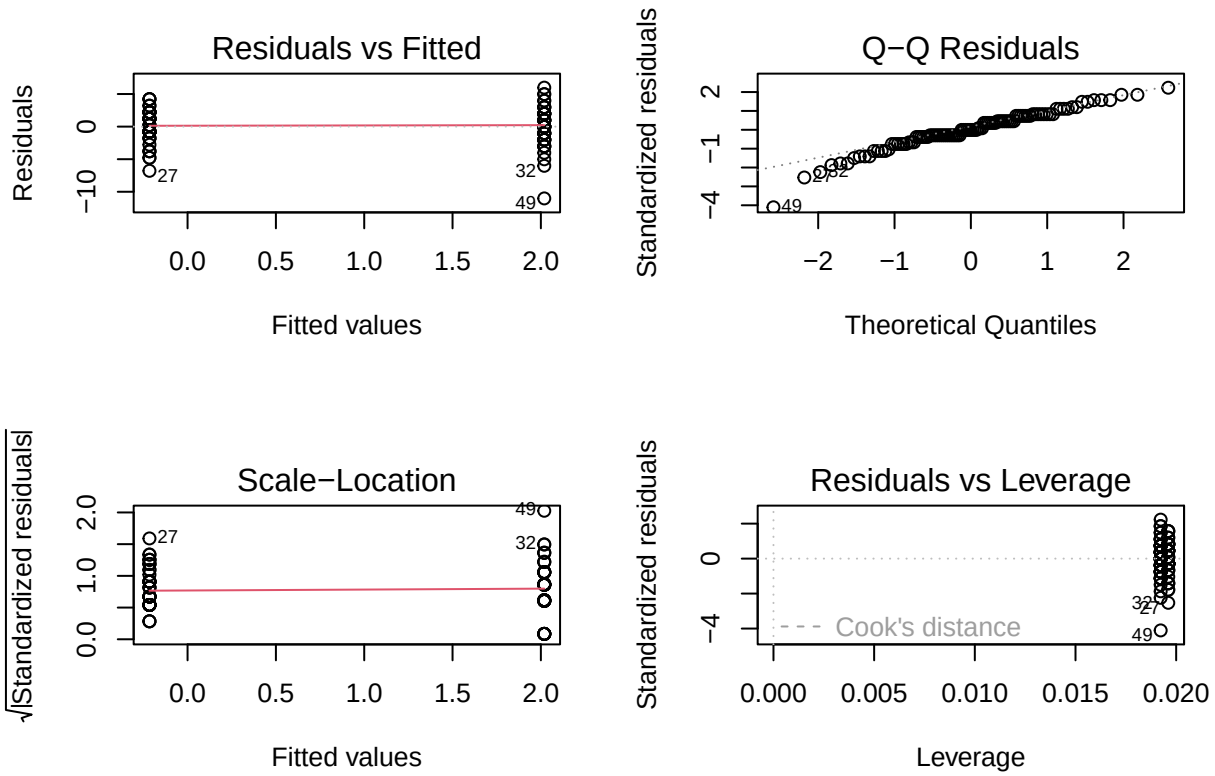
```
##
## Call:
## lm(formula = difference_s2_minus_s1 ~ Training_Condition, data = nep_wide_pre_post_wider)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.0192  -1.0192  -0.0192   1.9808   5.9808
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               -0.2157     0.3794  -0.568    0.571
## Training_ConditionTRAIN    2.2349     0.5340   4.185 6.09e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.71 on 101 degrees of freedom
## Multiple R-squared:  0.1478, Adjusted R-squared:  0.1393
## F-statistic: 17.51 on 1 and 101 DF,  p-value: 6.095e-05
```

1.3.3) 95% Confidence Intervals

```
##                              2.5 %    97.5 %
## (Intercept)             -0.9683904 0.5370178
## Training_ConditionTRAIN  1.1755628 3.2942713
```

```
##                             0.25 %   99.75 %
## (Intercept)             -1.3046783 0.8733057
## Training_ConditionTRAIN  0.7022718 3.7675623
```

1.3.4) Assumption tests



1.3.5) Linear Regression Model - adjusting for baseline instead of difference score (reviewer request)

```
lm_bp_prepost <- lm(zPost ~ Training_Condition + Apre, data = nep_wide_pre_post_wider)
```

1.3.6) Results

```
##
## Call:
## lm(formula = zPost ~ Training_Condition + Apre, data = nep_wide_pre_post_wider)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.0887 -1.3335  0.0991  1.7134  5.3698
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                4.7552     1.3831   3.438 0.000856 ***
## Training_ConditionTRAIN    2.3226     0.5036   4.612 1.18e-05 ***
## Apre                       0.3646     0.1708   2.135 0.035212 *
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.552 on 100 degrees of freedom
## Multiple R-squared:  0.2114, Adjusted R-squared:  0.1957
## F-statistic: 13.41 on 2 and 100 DF,  p-value: 6.948e-06
```

Standardised results:

```
## Parameter                   | Coefficient |   SE |         95% CI | t(100) |      p
## ------------------------------------------------------------------------------------
## (Intercept)                 |       -0.41 | 0.13 | [-0.66, -0.16] |  -3.28 | 0.001
## Training Condition [TRAIN] |        0.82 | 0.18 | [ 0.47,  1.17] |   4.61 | < .001
## Apre                        |        0.19 | 0.09 | [ 0.01,  0.37] |   2.13 | 0.035

##
## Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
##   using a Wald t-distribution approximation.

## Parameter                   | Coefficient |   SE |       99.5% CI | t(100) |      p
## ------------------------------------------------------------------------------------
## (Intercept)                 |       -0.41 | 0.13 | [-0.77, -0.05] |  -3.28 | 0.001
## Training Condition [TRAIN] |        0.82 | 0.18 | [ 0.31,  1.32] |   4.61 | < .001
## Apre                        |        0.19 | 0.09 | [-0.07,  0.44] |   2.13 | 0.035

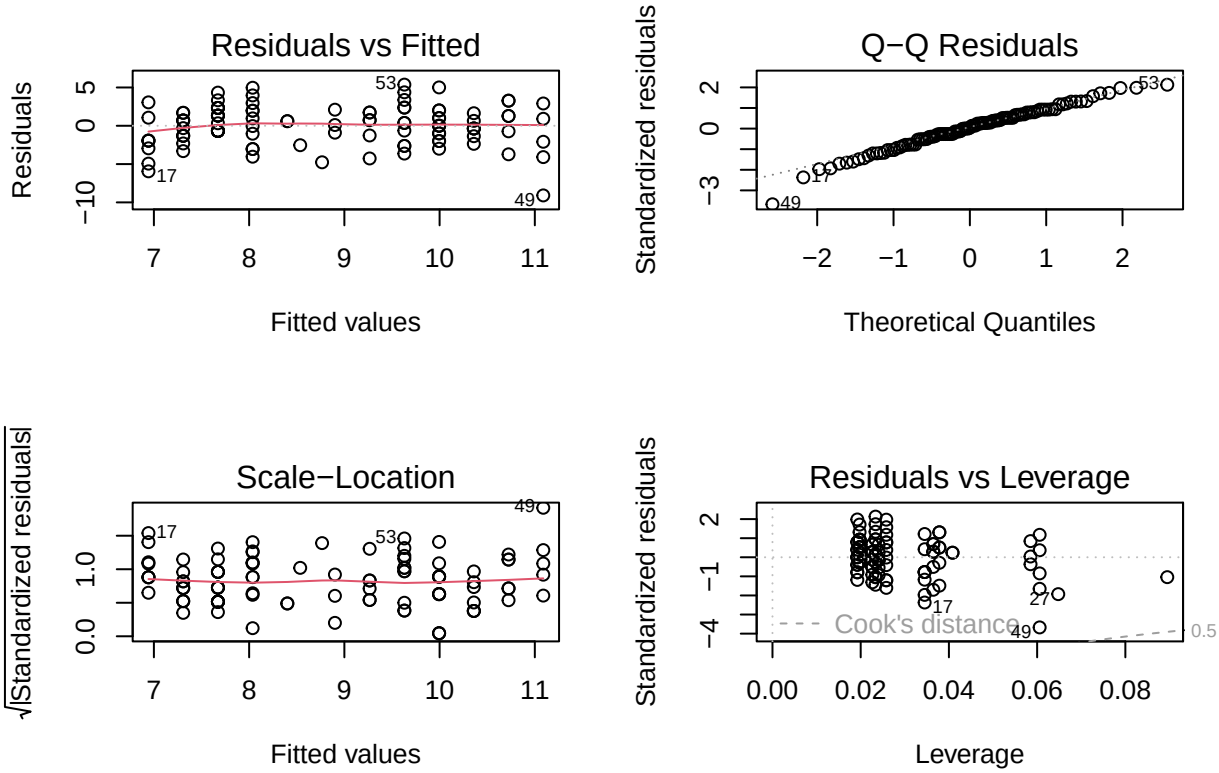
##
## Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
##   using a Wald t-distribution approximation.

## starting httpd help server ... done

##                               2.5 %    97.5 %
## (Intercept)             2.01110770 7.4993636
## Training_ConditionTRAIN 1.32356563 3.3216451
## Apre                    0.02577767 0.7034603

##                              0.25 %   99.75 %
## (Intercept)              0.7846972 8.7257741
## Training_ConditionTRAIN  0.8770730 3.7681377
## Apre                    -0.1256579 0.8548959
```

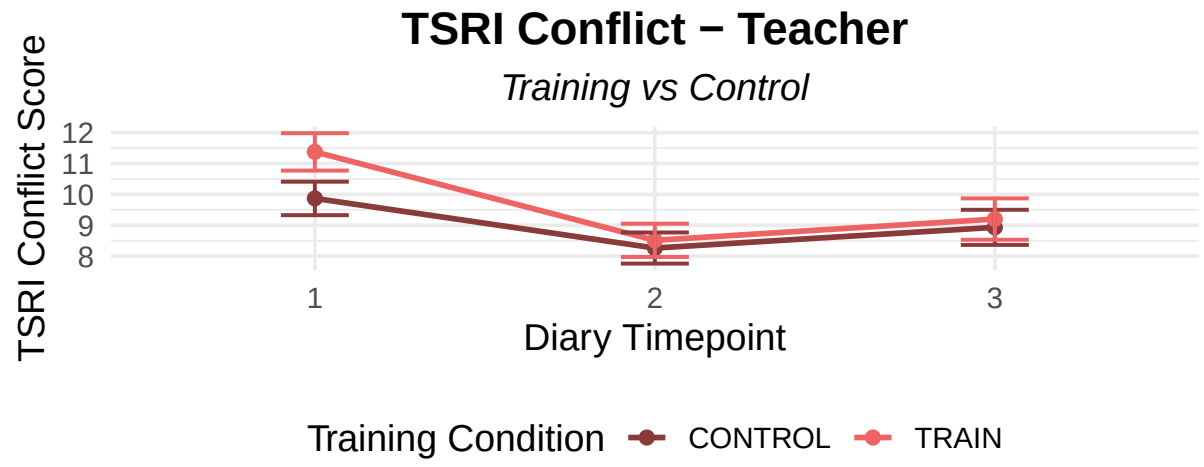
1.3.7) Assumption checks - pre post

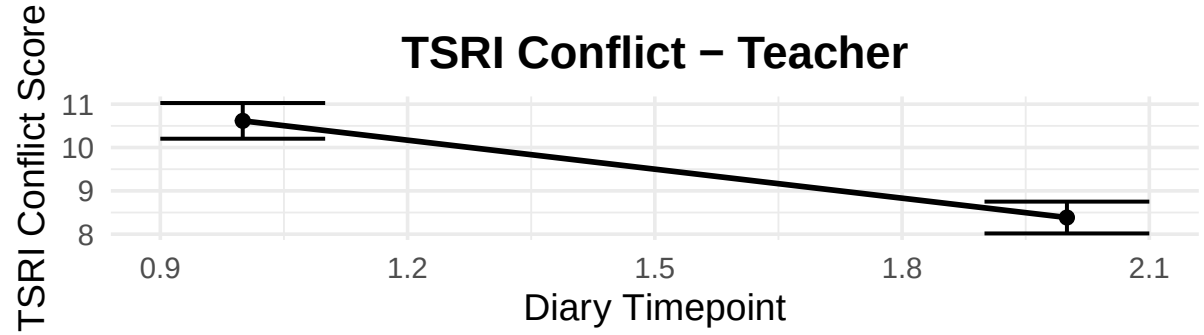


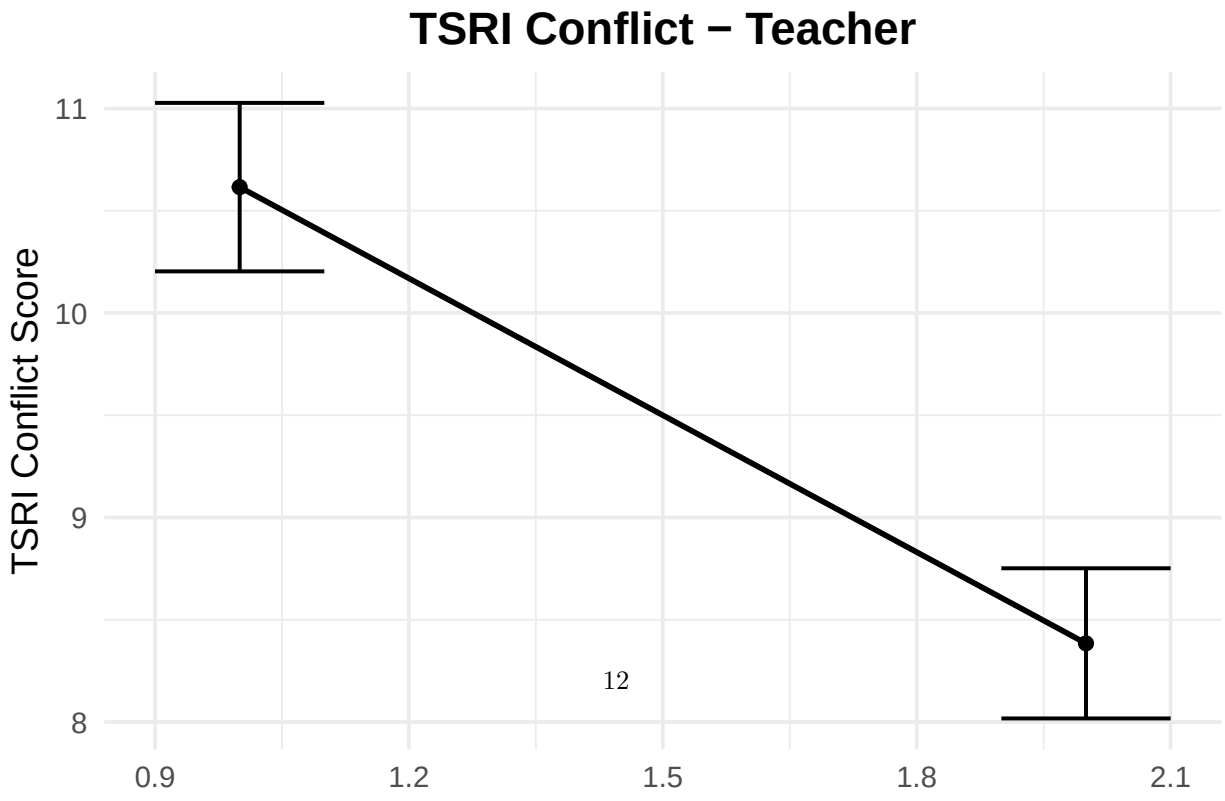
2) TSRI Conflict

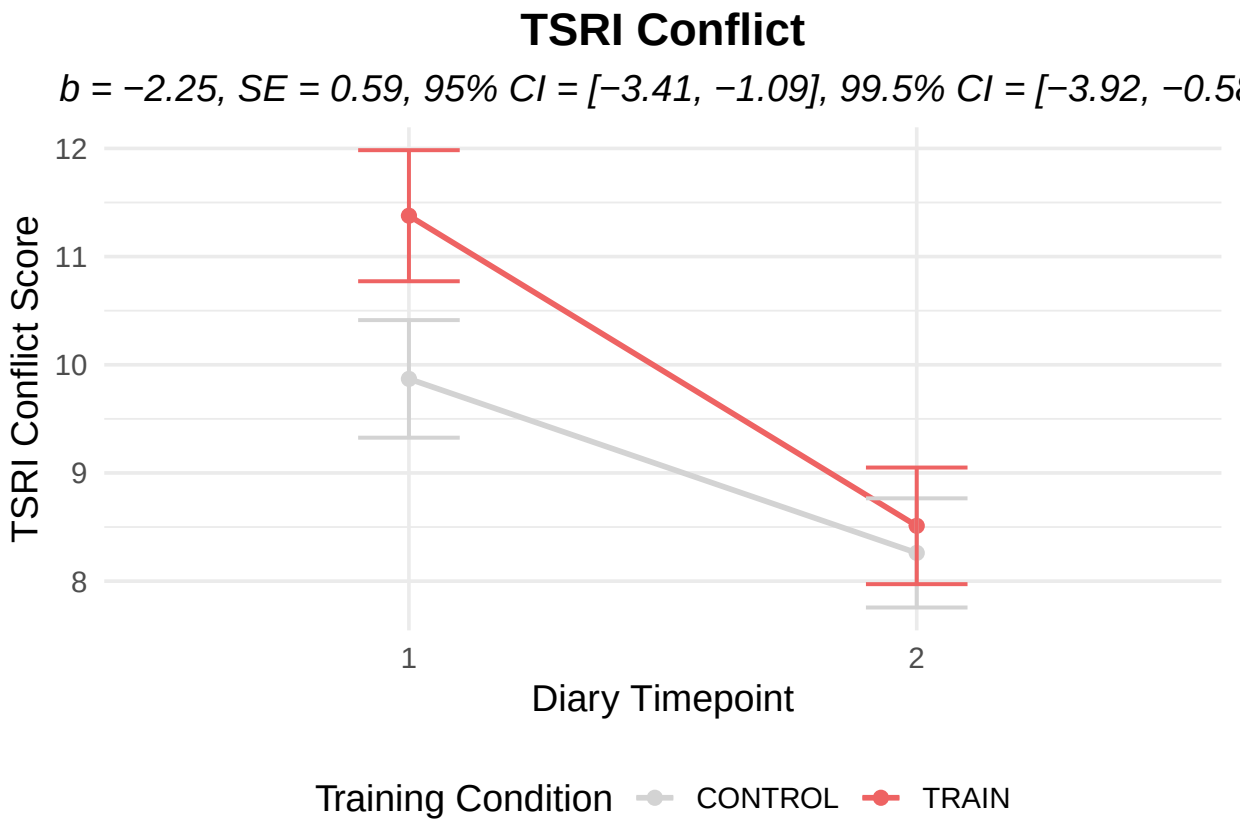
2.1) Summarising data: Do teacher ratings of Conflict at T2 differ by IBT condition?

2.1.1) Plot









2.1.2) Table

Training_Condition	n	M_diff_con_s2_s1	SE_diff_con_s2_s1
CONTROL	46	-1.608696	0.5187110
TRAIN	45	-2.866667	0.6192208

2.2) Model - Does IBT predict Conflict Difference Scores?

2.2.1) Linear Regression

```
model_conflict_t2 <- lm(diff_con_s2_s1 ~ Training_Condition, data = TSRI_con_wide)
```

LMM

```
model_conflict_t22 <- lm(TSRI_Conflict_T2 ~ Training_Condition + TSRI_Conflict_T1, data = TSRI_con_wide)
summary(model_conflict_t22)
```

```
##
## Call:
## lm(formula = TSRI_Conflict_T2 ~ Training_Condition + TSRI_Conflict_T1,
##     data = TSRI_con_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.8725 -2.2340 -0.4525  2.1660  7.8675
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               4.11563    0.96468   4.266 4.99e-05 ***
## Training_ConditionTRAIN -0.38321    0.67039  -0.572    0.569
## TSRI_Conflict_T1          0.42000    0.08577   4.897 4.39e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.137 on 88 degrees of freedom
## Multiple R-squared:  0.2151, Adjusted R-squared:  0.1973
## F-statistic: 12.06 on 2 and 88 DF,  p-value: 2.349e-05
```

2.2.2) Results

```
##
## Call:
## lm(formula = diff_con_s2_s1 ~ Training_Condition, data = TSRI_con_wide)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.1333  -2.1333   0.6087   2.1087   8.8667
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               -1.6087     0.5670  -2.837  0.00563 **
## Training_ConditionTRAIN   -1.2580     0.8063  -1.560  0.12226
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 3.846 on 89 degrees of freedom
## Multiple R-squared:  0.02662,    Adjusted R-squared:  0.01569
## F-statistic: 2.434 on 1 and 89 DF,  p-value: 0.1223
```

2.2.3) Confidence Intervals

```
##                               2.5 %     97.5 %
## (Intercept)             -2.735304 -0.4820871
## Training_ConditionTRAIN -2.860063  0.3441212
```

2.2.4) Linear Regression Model - adjusting for baseline instead of difference score (reviewer request)

```
lm_con_prepost <- lm(TSRI_Conflict_T2 ~ Training_Condition + TSRI_Conflict_T1, data = TSRI_con_wide)
```

2.2.5) Results

```
##
## Call:
## lm(formula = TSRI_Conflict_T2 ~ Training_Condition + TSRI_Conflict_T1,
##     data = TSRI_con_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.8725 -2.2340 -0.4525  2.1660  7.8675
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               4.11563    0.96468   4.266 4.99e-05 ***
## Training_ConditionTRAIN -0.38321    0.67039  -0.572    0.569
## TSRI_Conflict_T1          0.42000    0.08577   4.897 4.39e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.137 on 88 degrees of freedom
## Multiple R-squared:  0.2151, Adjusted R-squared:  0.1973
## F-statistic: 12.06 on 2 and 88 DF,  p-value: 2.349e-05
```

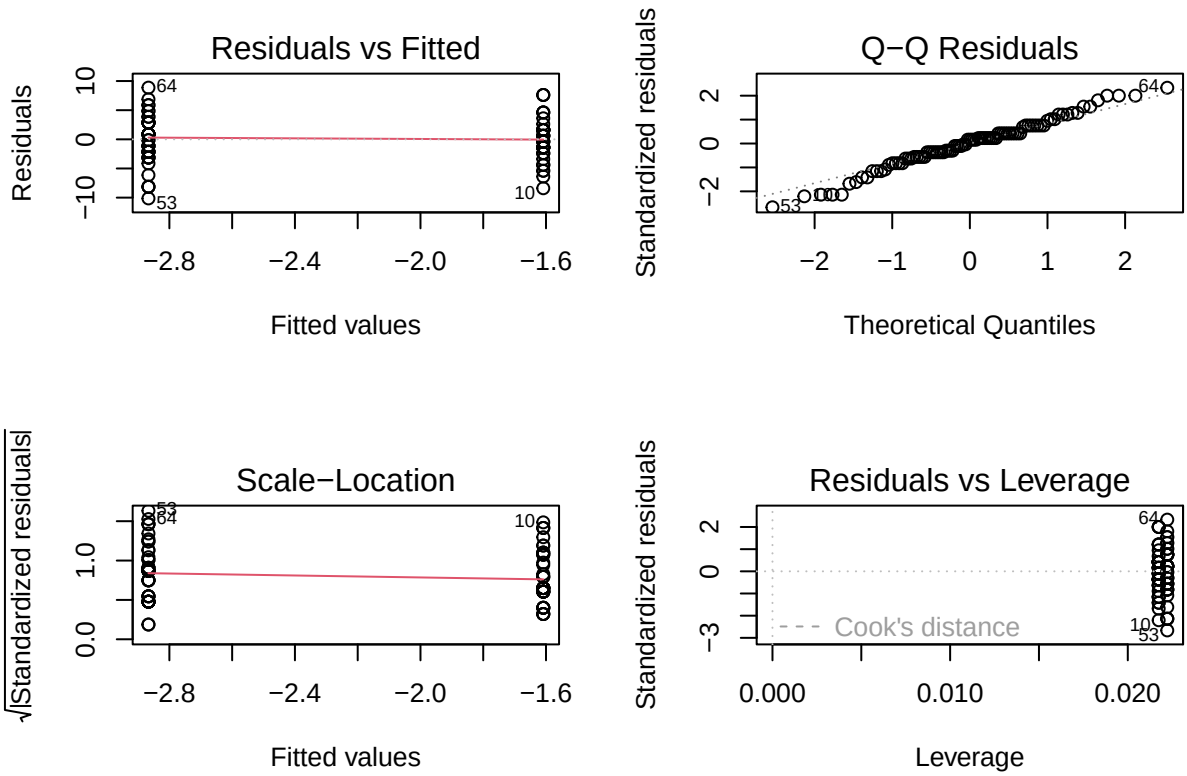
```
##                               2.5 %    97.5 %
## (Intercept)              2.1985344 6.0327352
## Training_ConditionTRAIN -1.7154617 0.9490409
## TSRI_Conflict_T1         0.2495446 0.5904590
```

Standardised results:

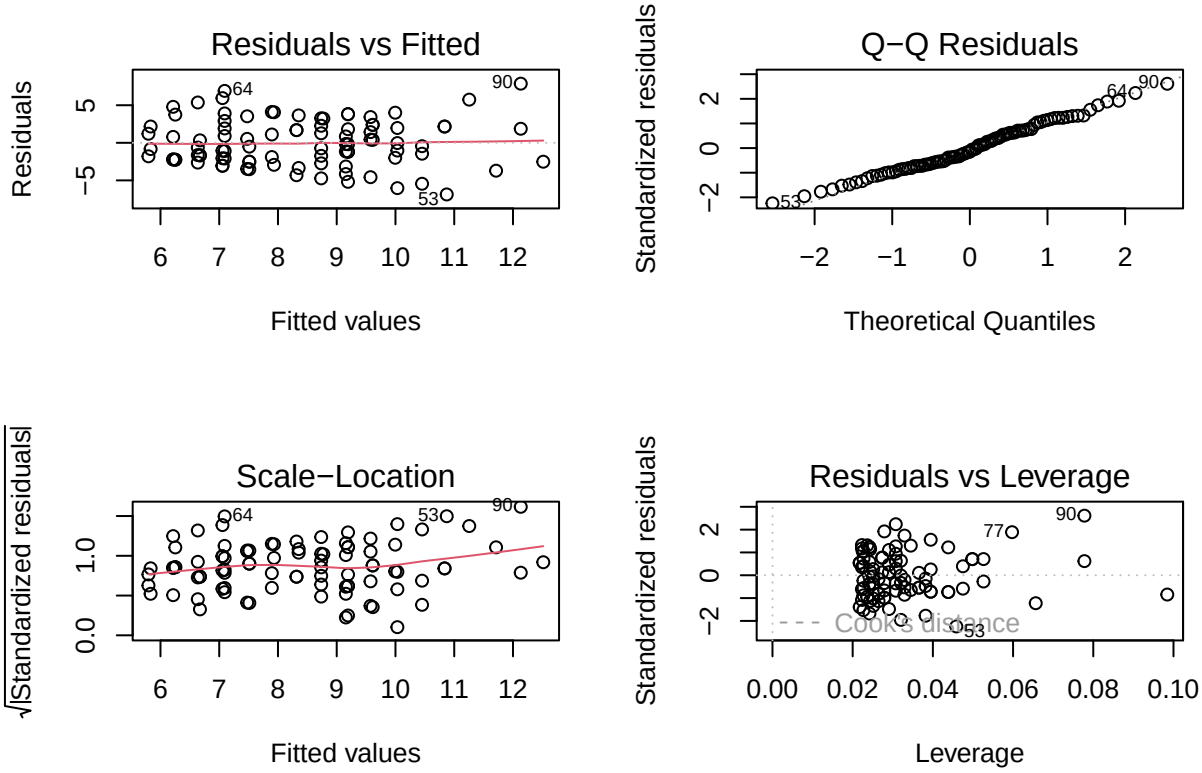
```
## Parameter                   | Coefficient |   SE |        95% CI | t(88) |      p
## ------------------------------------------------------------------------------------
## (Intercept)                 |        0.05 | 0.13 | [-0.21, 0.32] |  0.41 | 0.686
## Training Condition [TRAIN] |       -0.11 | 0.19 | [-0.49, 0.27] | -0.57 | 0.569
## TSRI Conflict T1            |        0.47 | 0.10 | [ 0.28, 0.66] |  4.90 | < .001
```

```
##
## Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
##   using a Wald t-distribution approximation.
```

2.2.6) Assumption Check - Difference model



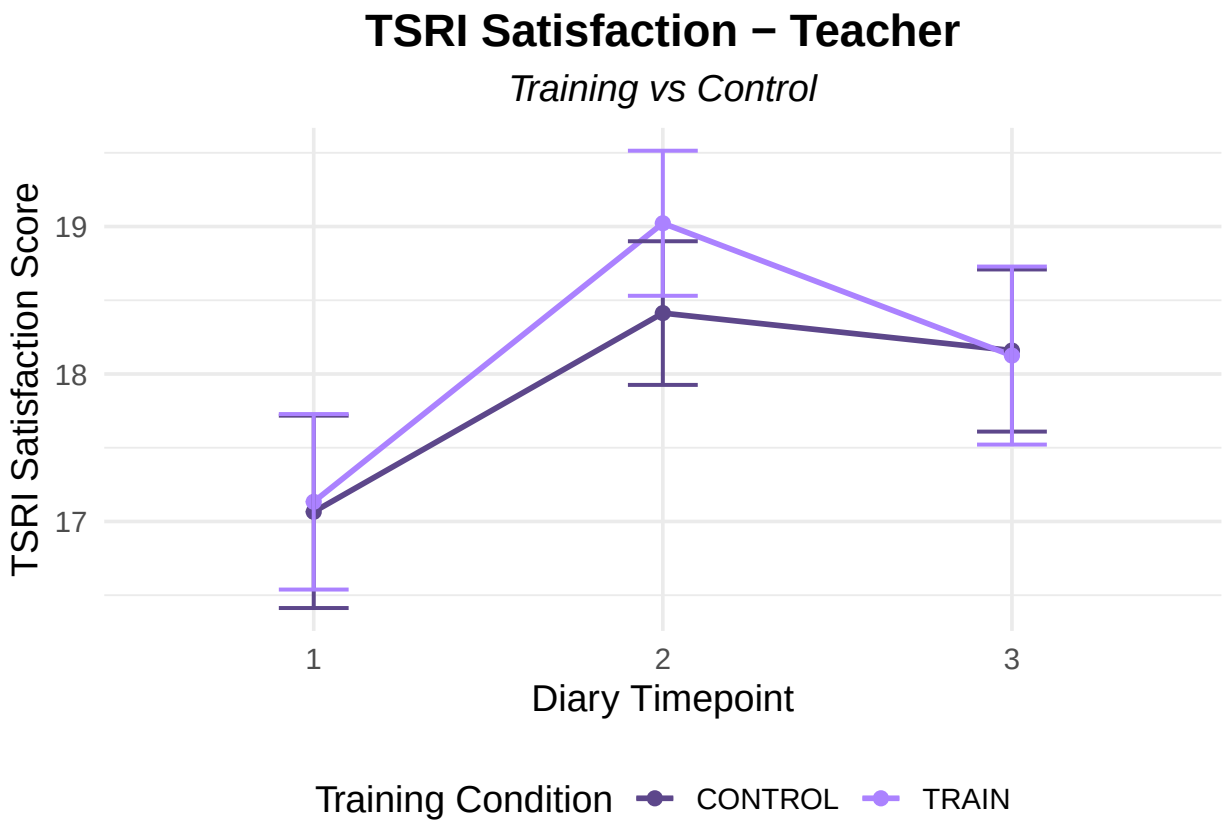
2.2.7) Assumption checks - pre post

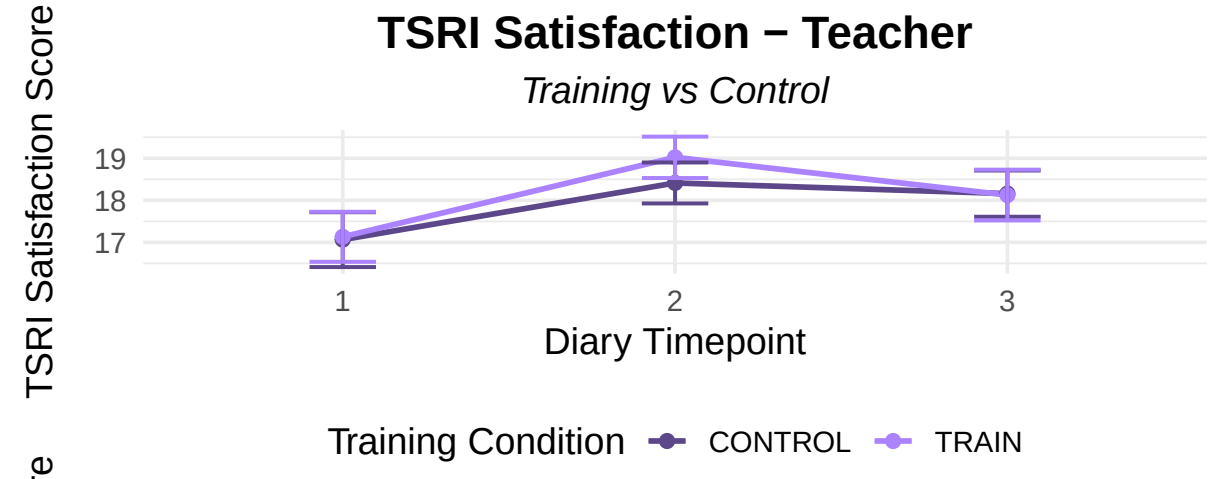

Residuals vs Fitted
Residuals
Fitted values
64
90
53
Q−Q Residuals
Standardized residuals
Theoretical Quantiles
64
90
53
Scale−Location
√|Standardized residuals|
Fitted values
64
53
90
Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance
77
90
53

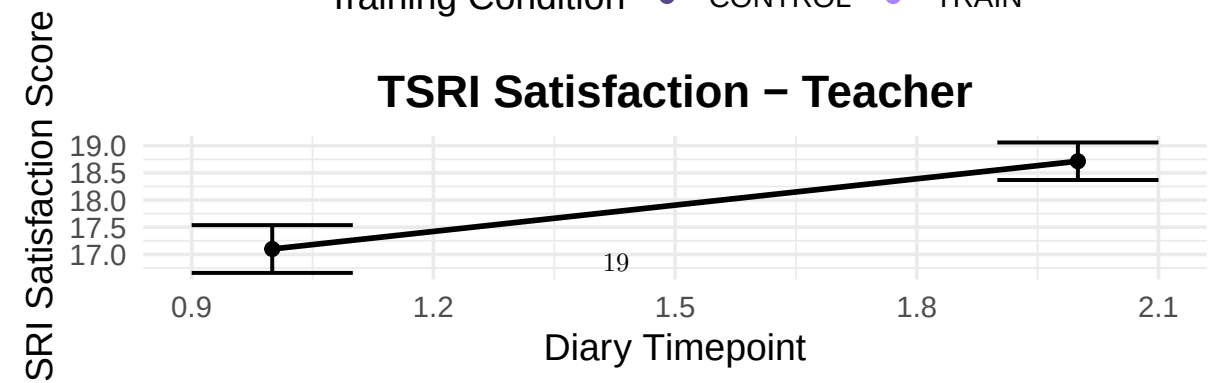
3) TSRI: Satisfaction

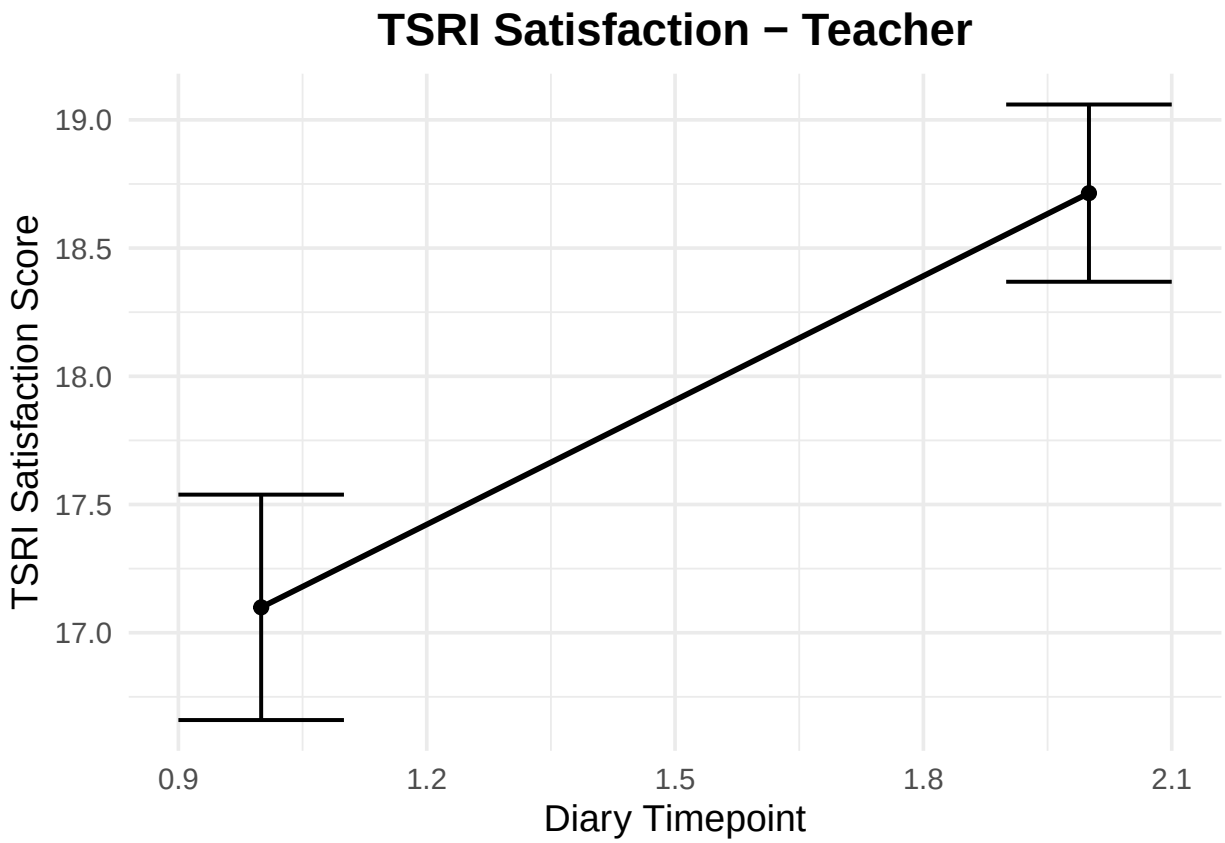
3.1) Summariseing Satisfaction Data

3.1.1) Plot: Do teacher ratings of Satisfaction T2 differ by IBT condition?










```
## Ignoring unknown labels:
## * colour : "Training Condition"
```



3.1.2) Table

Training_Condition	n	M_diff_Satisfaction_s2_s1	SE_diff_Satisfaction_s2_s1
CONTROL	46	1.347826	0.5403679
TRAIN	45	1.888889	0.4394926

3.2) Model - Do teacher ratings of Satisfaction T2 differ by IBT condition?

3.2.1) Linear Regression

```
model_satisfaction_t2 <- lm(diff_Satisfaction_s2_s1 ~ Training_Condition, data = TSRI_satisfaction_wide)
```

```
model_conflict_t22 <- lm(TSRI_Satisfaction_T2 ~ Training_Condition + TSRI_Satisfaction_T1, data = TSRI_s
summary(model_conflict_t22)
```

```
##
## Call:
## lm(formula = TSRI_Satisfaction_T2 ~ Training_Condition + TSRI_Satisfaction_T1,
##     data = TSRI_satisfaction_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.3596 -1.4370  0.0437  1.8156  7.6086
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               9.97077    1.16768   8.539 3.65e-13 ***
## Training_ConditionTRAIN   0.57548    0.53967   1.066    0.289
## TSRI_Satisfaction_T1      0.49471    0.06471   7.645 2.45e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.574 on 88 degrees of freedom
## Multiple R-squared:  0.4043, Adjusted R-squared:  0.3907
## F-statistic: 29.86 on 2 and 88 DF,  p-value: 1.265e-10
```

3.2.2) Results

```
##
## Call:
## lm(formula = diff_Satisfaction_s2_s1 ~ Training_Condition, data = TSRI_satisfaction_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.3478 -2.3478  0.1111  2.3816  8.6522
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               1.3478     0.4910   2.745  0.00732 **
## Training_ConditionTRAIN   0.5411     0.6982   0.775  0.44042
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.33 on 89 degrees of freedom
## Multiple R-squared:  0.006703,   Adjusted R-squared:  -0.004458
## F-statistic: 0.6006 on 1 and 89 DF,  p-value: 0.4404
```

3.2.3) 95% Confidence Intervalts

```
##                               2.5 %   97.5 %
## (Intercept)              0.3722721 2.323380
## Training_ConditionTRAIN -0.8462223 1.928348
```

3.2.4) Linear Regression Model - adjusting for baseline instead of difference score (reviewer request)

```
lm_sat_prepost <- lm(TSRI_Satisfaction_T2 ~ Training_Condition + TSRI_Satisfaction_T1, data = TSRI_satis
```

2.2.5) Results

```
##
## Call:
## lm(formula = TSRI_Satisfaction_T2 ~ Training_Condition + TSRI_Satisfaction_T1,
##     data = TSRI_satisfaction_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.3596 -1.4370  0.0437  1.8156  7.6086
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               9.97077    1.16768   8.539 3.65e-13 ***
## Training_ConditionTRAIN   0.57548    0.53967   1.066    0.289
## TSRI_Satisfaction_T1      0.49471    0.06471   7.645 2.45e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.574 on 88 degrees of freedom
## Multiple R-squared:  0.4043, Adjusted R-squared:  0.3907
## F-statistic: 29.86 on 2 and 88 DF,  p-value: 1.265e-10
```

```
##                               2.5 %     97.5 %
## (Intercept)               7.650249 12.2912964
## Training_ConditionTRAIN -0.496998  1.6479608
## TSRI_Satisfaction_T1      0.366108  0.6233046
```

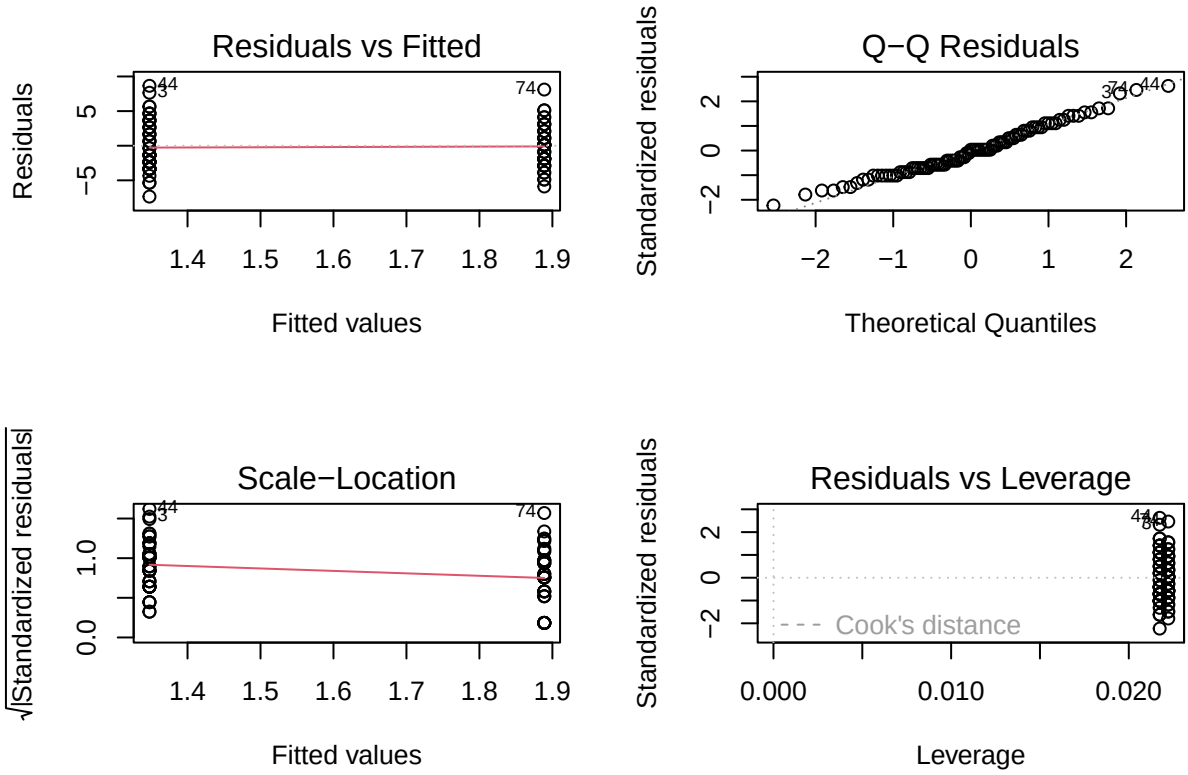
Standardised results:

```
## Parameter                   | Coefficient |   SE |        95% CI | t(88) |      p
## ------------------------------------------------------------------------------------
## (Intercept)                 |       -0.09 | 0.12 | [-0.32, 0.14] | -0.75 | 0.455
## Training Condition [TRAIN] |        0.17 | 0.16 | [-0.15, 0.50] |  1.07 | 0.289
## TSRI Satisfaction T1        |        0.63 | 0.08 | [ 0.47, 0.79] |  7.64 | < .001

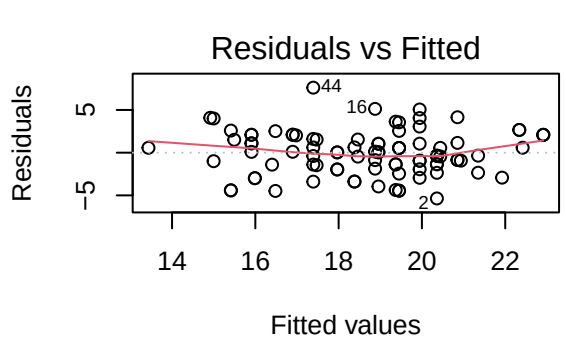
##
## Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
##   using a Wald t-distribution approximation.
```

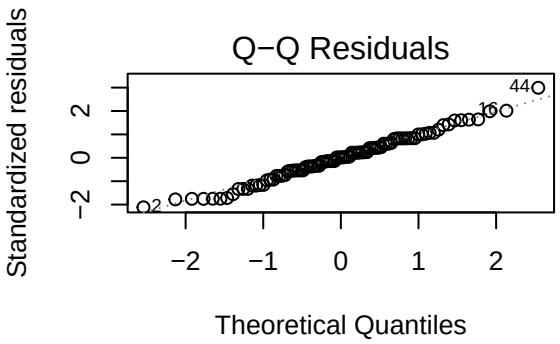
2.2.6) Assumption Checks

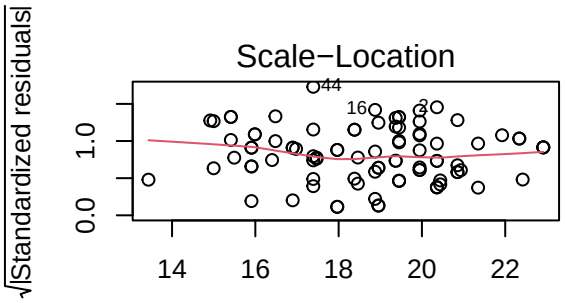
Difference model

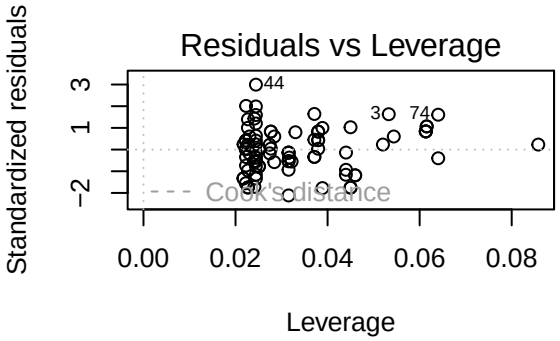


Model adjusting for baseline


Residuals vs Fitted
Residuals
Fitted values
44
16
2


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
44
16
2

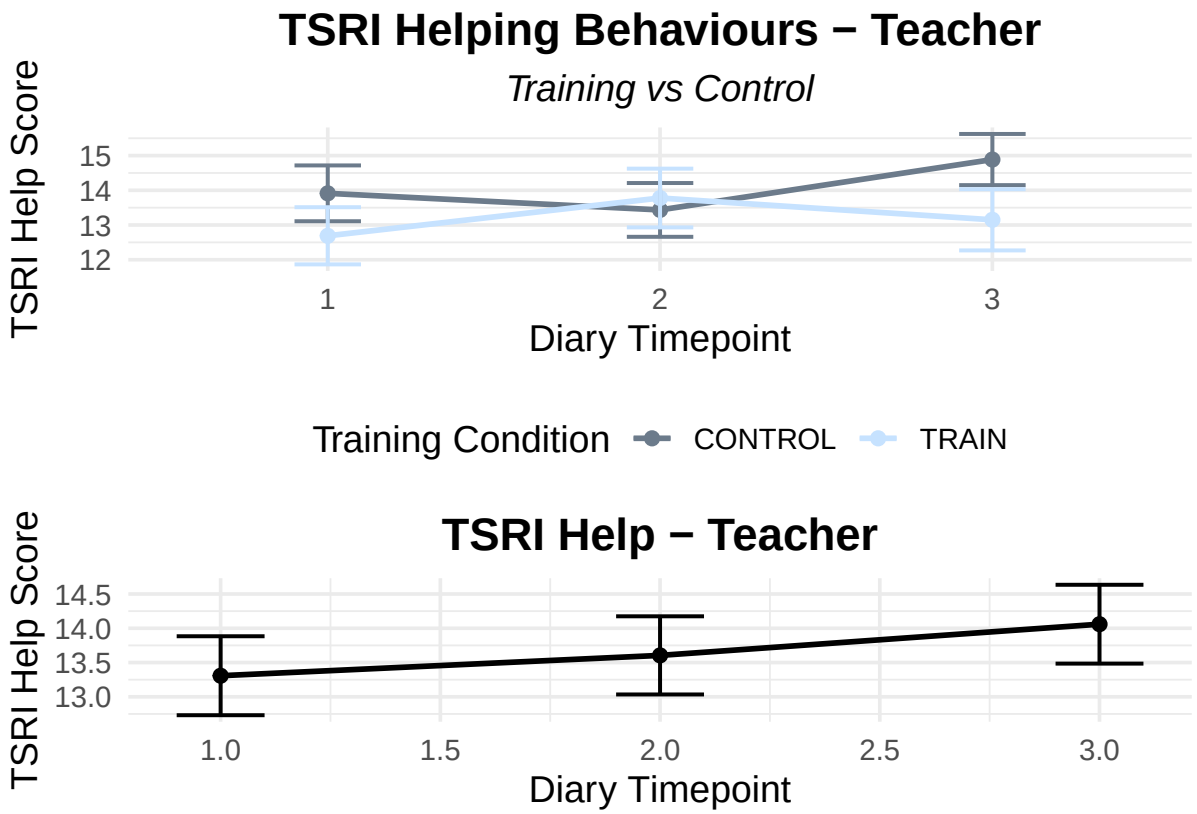

Scale−Location
√|Standardized residuals|
Fitted values
44
16
3


Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance
44
3
74

4) TSRI: Helping behaviours

4.1) Summarising data: Do teacher ratings of Help at T2 differ by IBT condition?

4.1.1) Plot:



4.1.2) Table

Training_Condition	n	M_diff_Help_s2_s1	SE_diff_Help_s2_s1
CONTROL	46	-0.4782609	0.4762364
TRAIN	45	1.0888889	0.5841529

4.2) Model: Do teacher ratings of Help at T2 differ by IBT condition?

4.2.1) Linear Regression

```
model_help_t2 <- lm(diff_Help_s2_s1 ~ Training_Condition, data = TSRI_help_wide)
```

```
model_conflict_t22 <- lm(TSRI_Help_T2 ~ Training_Condition + TSRI_Help_T1, data = TSRI_help_wide)
summary(model_conflict_t22)
```

```
##
## Call:
## lm(formula = TSRI_Help_T2 ~ Training_Condition + TSRI_Help_T1,
##     data = TSRI_help_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.0212 -1.6765 -0.5452  2.1102  9.7147
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               2.54809    1.04220   2.445   0.0165 *
## Training_ConditionTRAIN   1.30087    0.71770   1.813   0.0733 .
## TSRI_Help_T1              0.78248    0.06566  11.916   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.401 on 88 degrees of freedom
## Multiple R-squared:  0.6178, Adjusted R-squared:  0.6091
## F-statistic: 71.11 on 2 and 88 DF,  p-value: < 2.2e-16
```

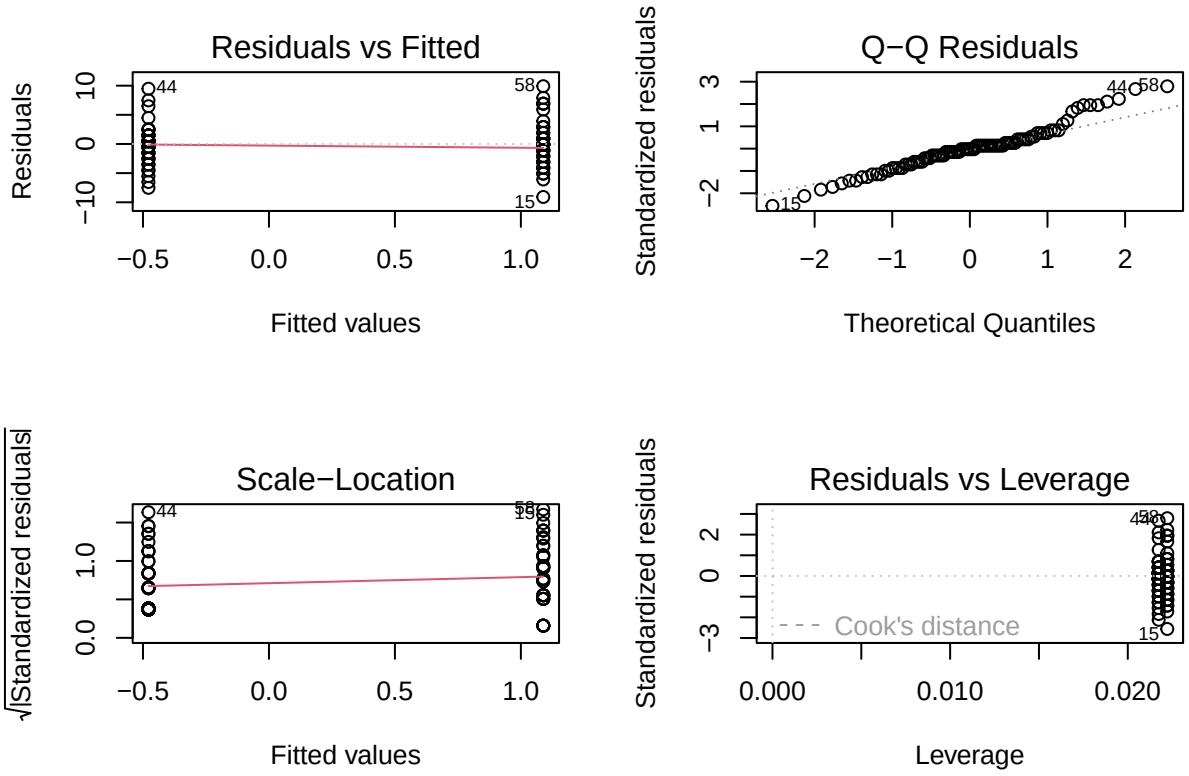
4.2.2) Results

```
##
## Call:
## lm(formula = diff_Help_s2_s1 ~ Training_Condition, data = TSRI_help_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.0889 -2.0889 -0.0889  1.4783  9.9111
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               -0.4783     0.5289  -0.904   0.3683
## Training_ConditionTRAIN    1.5671     0.7521   2.084   0.0401 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.587 on 89 degrees of freedom
## Multiple R-squared:  0.04652,    Adjusted R-squared:  0.0358
## F-statistic: 4.342 on 1 and 89 DF,  p-value: 0.04005
```

4.2.3) Confidence Intervals

```
##                                2.5 %    97.5 %
## (Intercept)              -1.52912194 0.5726002
## Training_ConditionTRAIN   0.07277424 3.0615253
```

```
##                               0.25 %  99.75 %
## (Intercept)              -2.0007295 1.044208
## Training_ConditionTRAIN -0.5978746 3.732174
```



3.2.4) Linear Regression Model - adjusting for baseline instead of difference score (reviewer request)

```
lm_hel_prepost <- lm(TSRI_Help_T2 ~ Training_Condition + TSRI_Help_T1, data = TSRI_help_wide)
```

2.2.5) Results

```
##
## Call:
## lm(formula = TSRI_Help_T2 ~ Training_Condition + TSRI_Help_T1,
##     data = TSRI_help_wide)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.0212 -1.6765 -0.5452  2.1102  9.7147
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               2.54809    1.04220   2.445   0.0165 *
## Training_ConditionTRAIN   1.30087    0.71770   1.813   0.0733 .
## TSRI_Help_T1              0.78248    0.06566  11.916   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.401 on 88 degrees of freedom
## Multiple R-squared:  0.6178, Adjusted R-squared:  0.6091
## F-statistic: 71.11 on 2 and 88 DF,  p-value: < 2.2e-16
```

```
##                               2.5 %    97.5 %
## (Intercept)               0.4769368 4.6192504
## Training_ConditionTRAIN -0.1254094 2.7271546
## TSRI_Help_T1              0.6519856 0.9129759
```

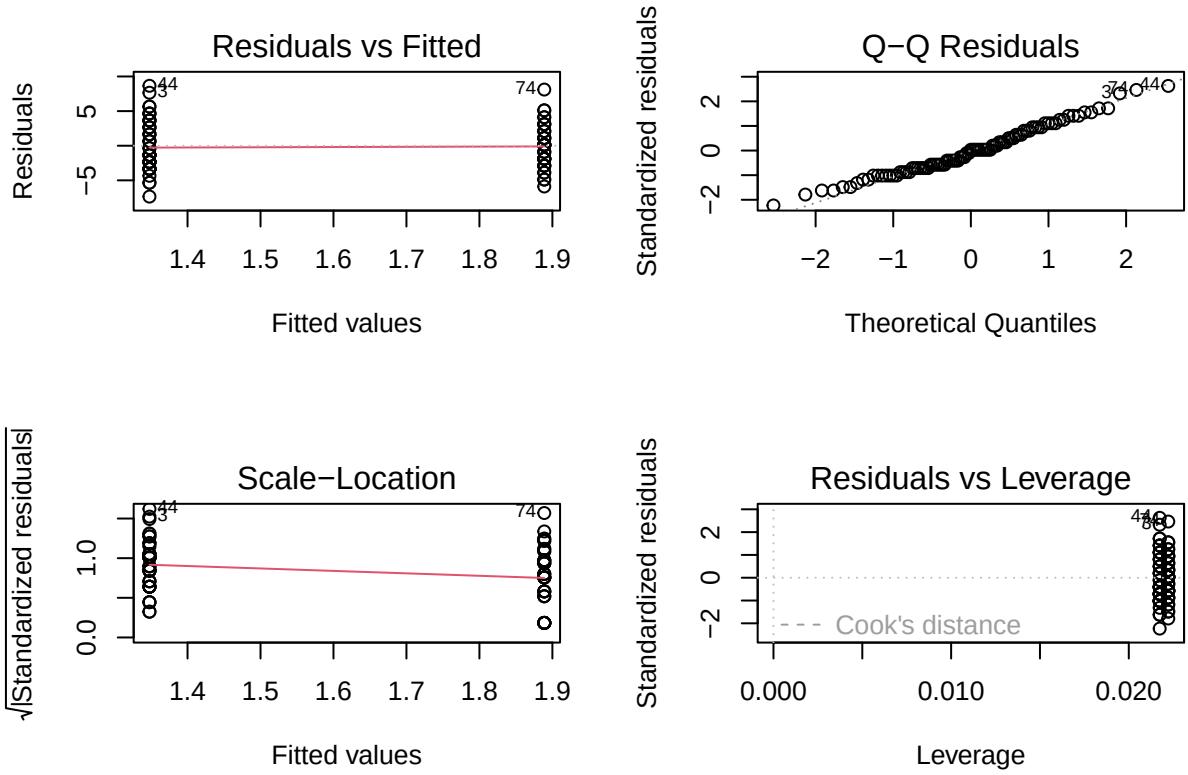
Standardised results:

```
## Parameter                   | Coefficient |   SE |        95% CI | t(88) |      p
## ------------------------------------------------------------------------------------
## (Intercept)                 |       -0.12 | 0.09 | [-0.30, 0.07] | -1.28 | 0.204
## Training Condition [TRAIN] |        0.24 | 0.13 | [-0.02, 0.50] |  1.81 | 0.073
## TSRI Help T1                |        0.79 | 0.07 | [ 0.66, 0.92] | 11.92 | < .001
```

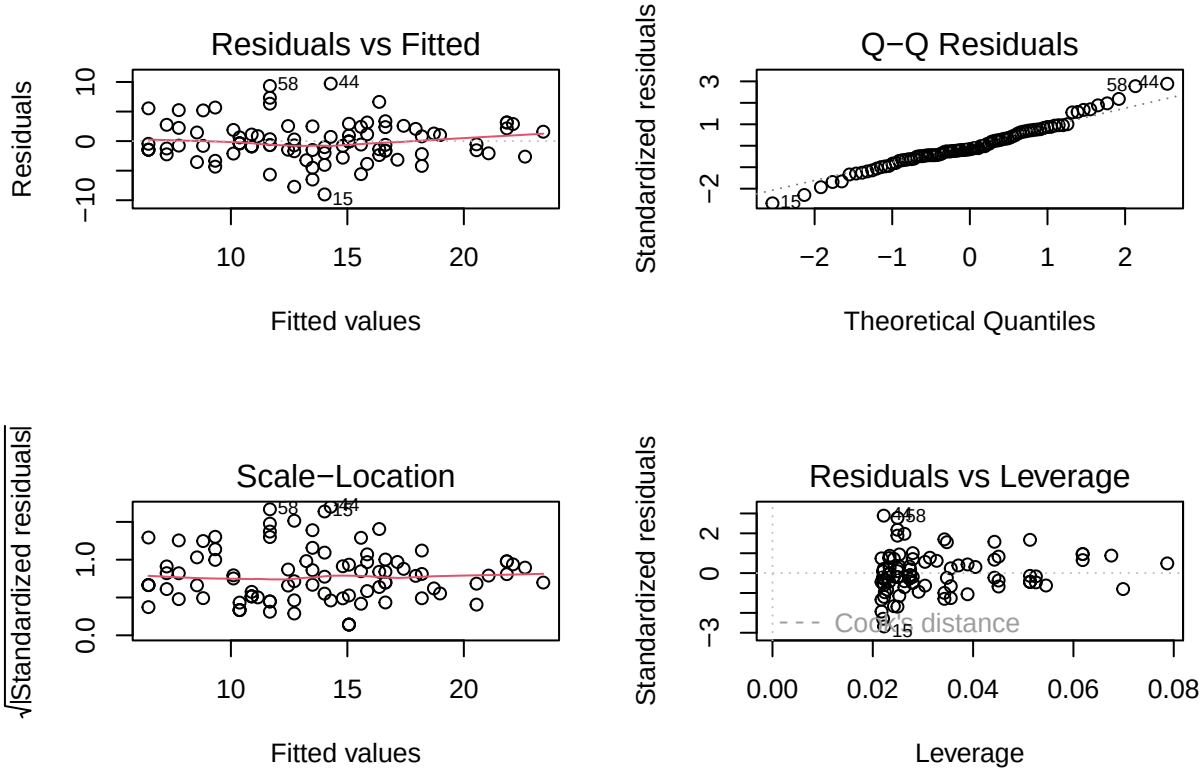
```
##
## Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
##   using a Wald t-distribution approximation.
```

4.2.4) Assumption checks

Difference model



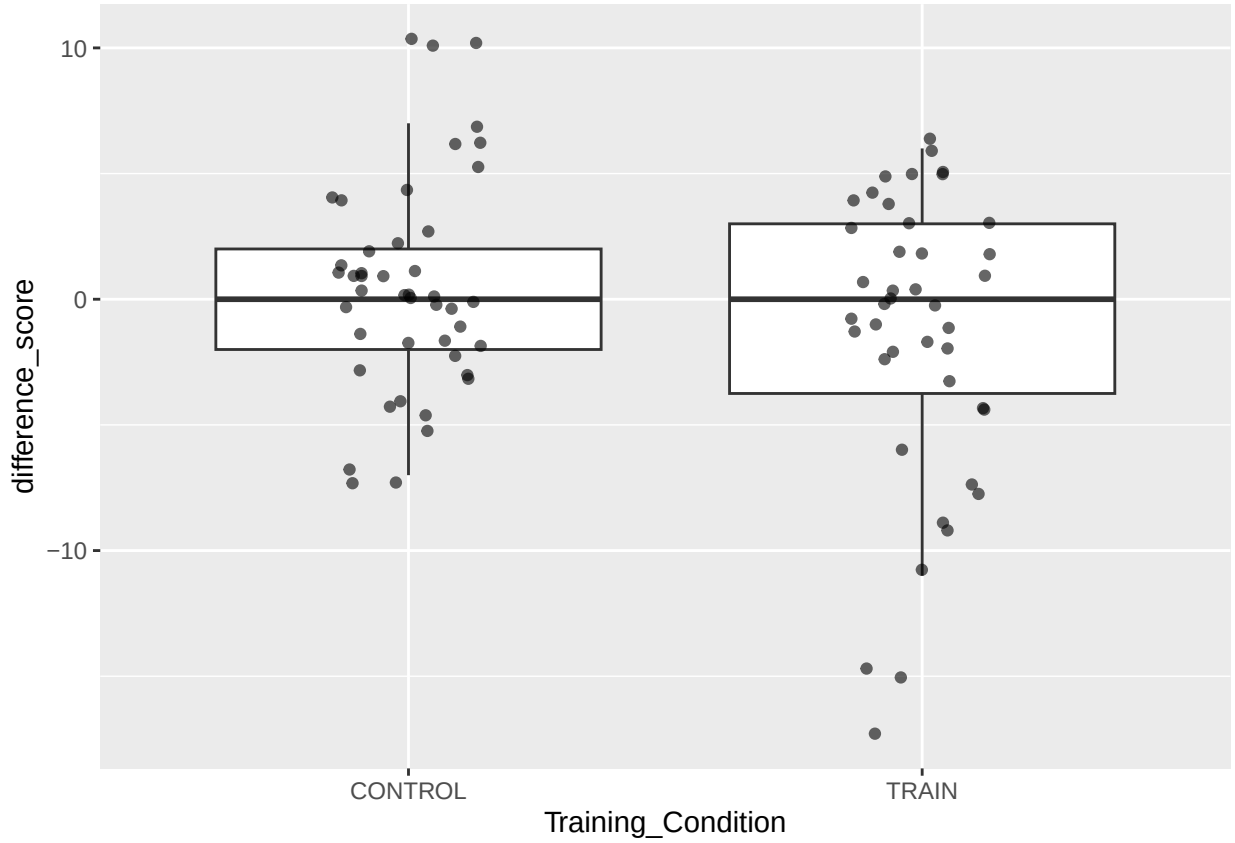
Model adjusting for basline


Residuals vs Fitted
Residuals
Fitted values
Q−Q Residuals
Standardized residuals
Theoretical Quantiles
Scale−Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage

5) Child SRQ-A Prosocial Intentions.

5.1) Sumarising child SRQ data

5.1.1) Plotting child SRQ difference scores



5.2.2) Table

Training_Condition	n	M_diff_difference_score	SE_diff_difference_score
CONTROL	45	0.4888889	0.6281464
TRAIN	42	-1.4047619	0.9049224

5.2) Model: child SRQ Difference Scores

5.2.1) Linear Regression

```
# S2 - S1 so we want the score to be positive
model_srq_diff <- lm(difference_score ~ Training_Condition, data = child_srq)
```

5.2.2) Results

```
##
## Call:
## lm(formula = difference_score ~ Training_Condition, data = child_srq)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.5952  -2.4889   0.4048   3.4579   9.5111
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                0.4889     0.7569   0.646   0.5201
## Training_ConditionTRAIN   -1.8937     1.0894  -1.738   0.0858 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.077 on 85 degrees of freedom
## Multiple R-squared:  0.03433,    Adjusted R-squared:  0.02297
## F-statistic: 3.022 on 1 and 85 DF,  p-value: 0.08578
```

5.2.3) 95% Confidence Intervals

```
##                              2.5 %    97.5 %
## (Intercept)             -1.016041 1.9938192
## Training_ConditionTRAIN -4.059615 0.2723139
```

5.2.4) Linear Regression Model - adjusting for baseline instead of difference score (reviewer request)

```
lm_srq_prepost <- lm(SRQ_Prosocial_S2 ~ Training_Condition + SRQ_Prosocial_S1, data = child_srq)
```

2.2.5) Results

```
##
## Call:
## lm(formula = SRQ_Prosocial_S2 ~ Training_Condition + SRQ_Prosocial_S1,
##     data = child_srq)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.8636  -2.1030   0.7738   2.6869   7.0355
```

```
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               16.0408     2.7647   5.802 1.13e-07 ***
## Training_ConditionTRAIN  -1.9424     0.9268  -2.096   0.0391 *
## SRQ_Prosocial_S1          0.4202     0.1002   4.192 6.82e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.319 on 84 degrees of freedom
## Multiple R-squared:  0.2085, Adjusted R-squared:  0.1896
## F-statistic: 11.06 on 2 and 84 DF,  p-value: 5.437e-05
```

Confidence Intervals:

```
##                              2.5 %      97.5 %
## (Intercept)              10.5428650 21.53878370
## Training_ConditionTRAIN -3.7853827 -0.09947516
## SRQ_Prosocial_S1         0.2208434  0.61952595
```

```
##                              0.25 %    99.75 %
## (Intercept)               8.0699256 24.0117231
## Training_ConditionTRAIN -4.6143289  0.7294711
## SRQ_Prosocial_S1         0.1311812  0.7091881
```

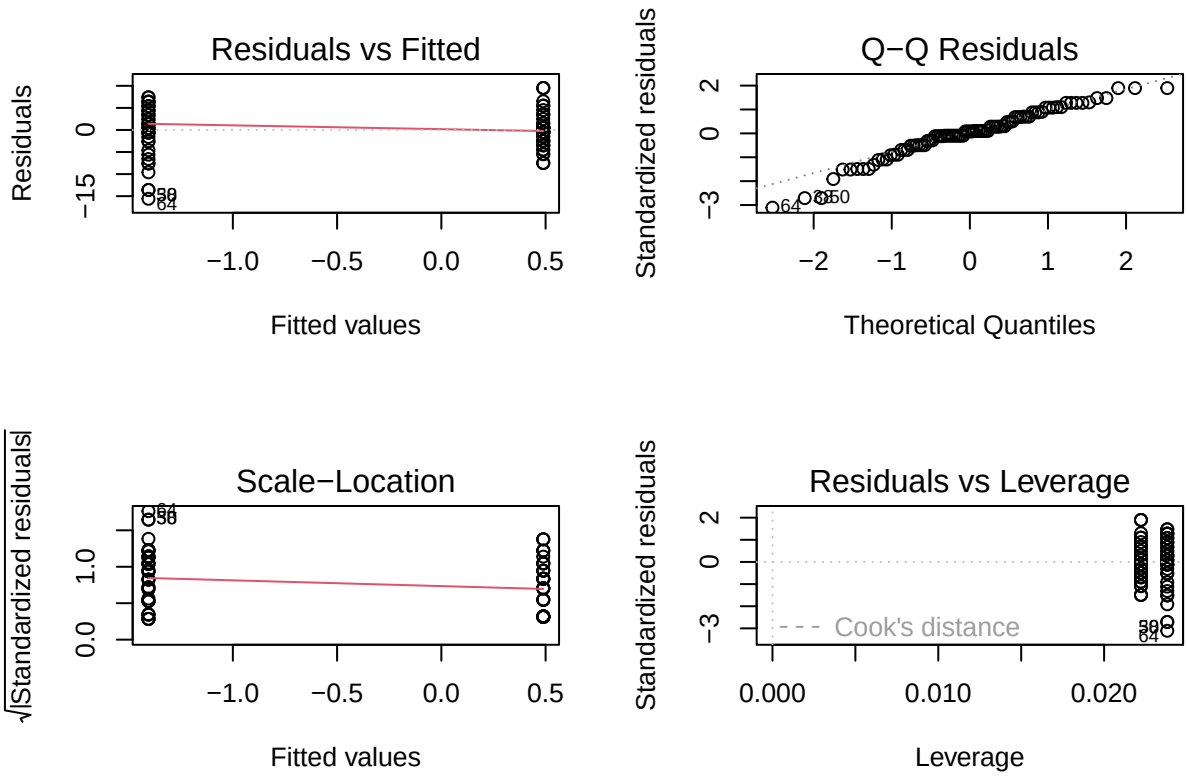
Standardised results:

```
## Parameter                   | Coefficient |   SE |         95% CI | t(84) |      p
## ------------------------------------------------------------------------------------
## (Intercept)                 |        0.20 | 0.13 | [-0.07,  0.46] |  1.46 | 0.149
## Training Condition [TRAIN] |       -0.40 | 0.19 | [-0.79, -0.02] | -2.10 | 0.039
## SRQ Prosocial S1            |        0.41 | 0.10 | [ 0.21,  0.60] |  4.19 | < .001
```

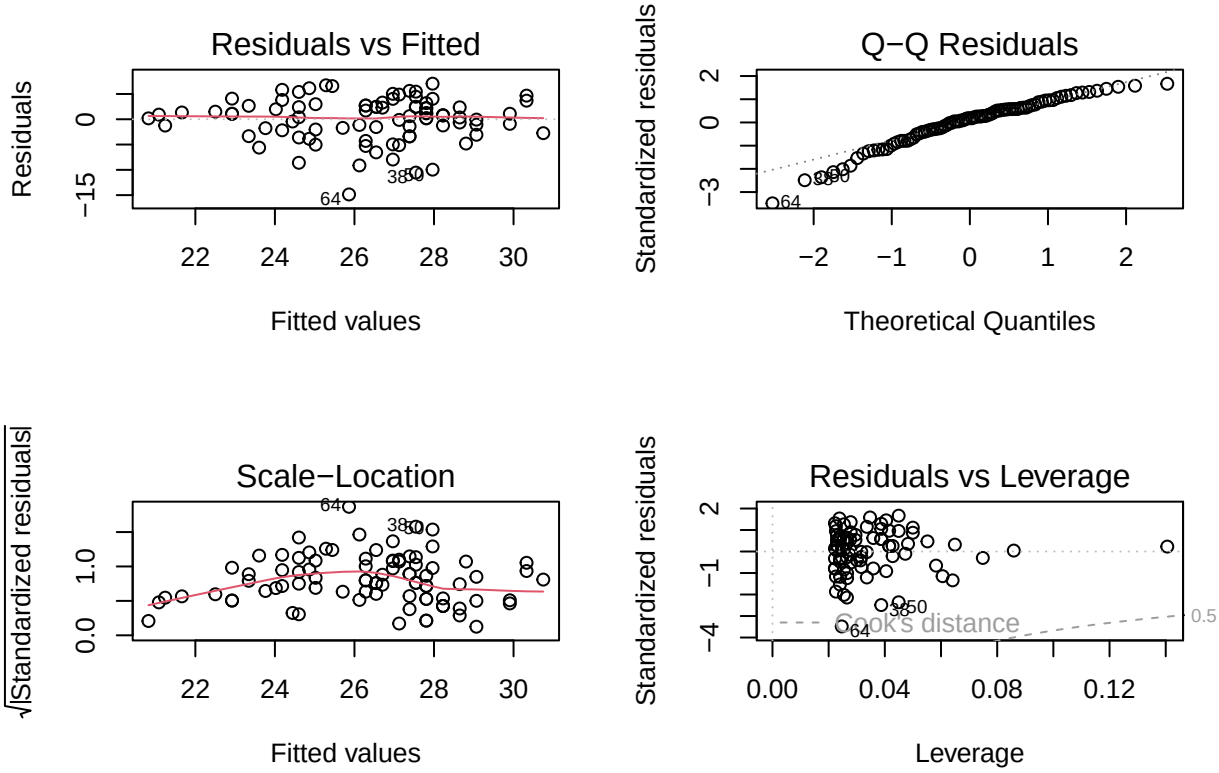
```
##
## Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
##   using a Wald t-distribution approximation.
```

```
## # Standardization method: refit
##
## Parameter                   | Std. Coef. |      99.5% CI
## ------------------------------------------------------
## (Intercept)                 |       0.20 | [-0.19, 0.58]
## Training Condition [TRAIN] |      -0.40 | [-0.96, 0.15]
## SRQ Prosocial S1            |       0.41 | [ 0.13, 0.69]
```

Assumption Check: difference model



Assuption check: mdel adjusting for baseline


Residuals vs Fitted
Residuals
Fitted values
Q−Q Residuals
Standardized residuals
Theoretical Quantiles
Scale−Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage

6) Exploratory analyses

6.1) (exploratory) Do teacher ratings of conflict differ from T1-3

Number of participants:

```
length(unique(TSRI_ICU_d1d2d3_full$ID_numeric))
```

```
## [1] 84
```

```
model_conflict_exploratory_simpler <- lm(T_TSRI_Conflict_Total ~ diaryTP , data = TSRI_ICU_d1d2d3_full)


summary(model_conflict_exploratory_simpler)
```

```
##
## Call:
## lm(formula = T_TSRI_Conflict_Total ~ diaryTP, data = TSRI_ICU_d1d2d3_full)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.4286 -3.1786 -0.1786  2.9405 11.8214
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  10.4286     0.4179  24.952  < 2e-16 ***
## diaryTP2     -2.2500     0.5911  -3.807 0.000177 ***
## diaryTP3     -1.3690     0.5911  -2.316 0.021358 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.831 on 249 degrees of freedom
## Multiple R-squared:  0.05581,    Adjusted R-squared:  0.04823
## F-statistic: 7.359 on 2 and 249 DF,  p-value: 0.000785
```

```
confint(model_conflict_exploratory_simpler)
```

```
##                 2.5 %     97.5 %
## (Intercept)  9.605410 11.2517329
## diaryTP2    -3.414126 -1.0858739
## diaryTP3    -2.533174 -0.2049216
```

```
confint(model_conflict_exploratory_simpler, level = 0.995)
```

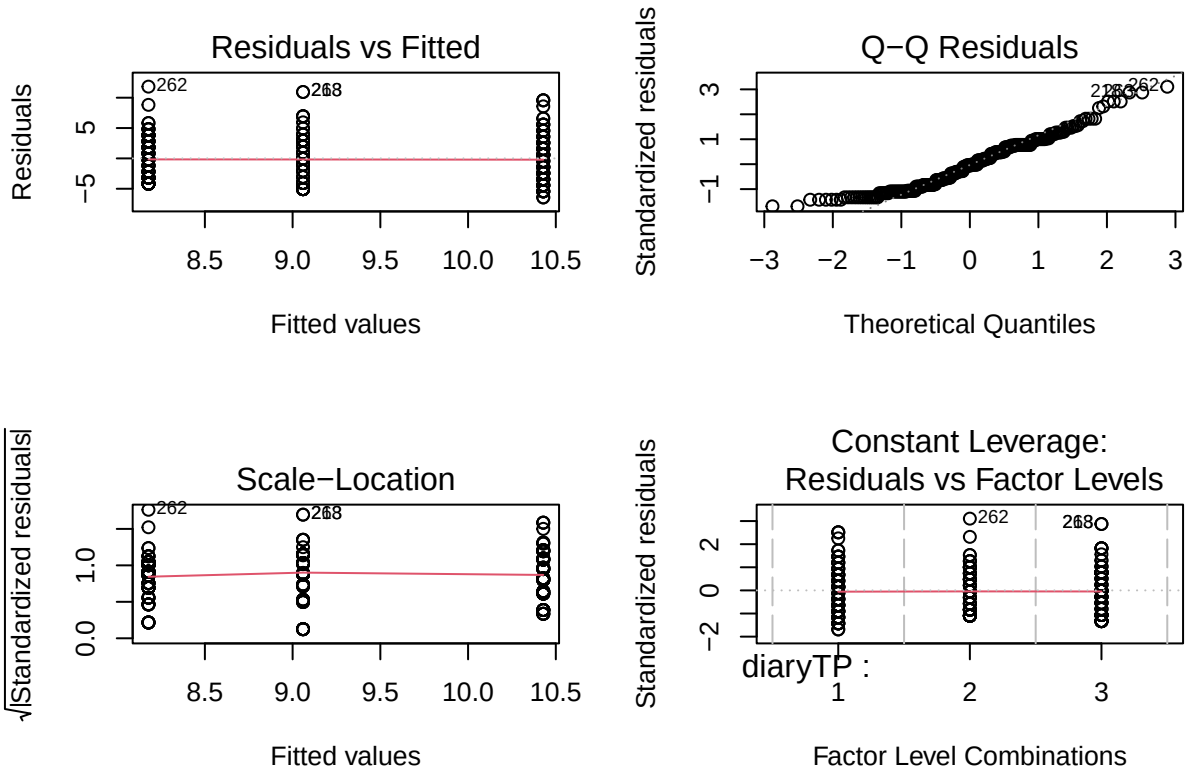
```
##                0.25 %    99.75 %
## (Intercept)  9.244835 11.6123082
## diaryTP2    -3.924057 -0.5759434
## diaryTP3    -3.043104  0.3050090
```

Standardised parameters:

```
## Parameter   | Coefficient |   SE |         95% CI | t(249) |      p
## ------------------------------------------------------------------
## (Intercept) |        0.31 | 0.11 | [ 0.10,  0.52] |   2.89 | 0.004
## diaryTP [2] |       -0.57 | 0.15 | [-0.87, -0.28] |  -3.81 | < .001
## diaryTP [3] |       -0.35 | 0.15 | [-0.65, -0.05] |  -2.32 | 0.021

##
## Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
##   using a Wald t-distribution approximation.

## # Standardization method: refit
##
## Parameter   | Std. Coef. |       99.5% CI
## ----------------------------------------
## (Intercept) |       0.31 | [ 0.01,  0.61]
## diaryTP [2] |      -0.57 | [-1.00, -0.15]
## diaryTP [3] |      -0.35 | [-0.78,  0.08]
```



6.2) (exploratory) Do teacher ratings of satisfaction differ from T1-3

```
##
## Call:
## lm(formula = T_TSRI_Satisfaction_Total ~ diaryTP, data = TSRI_ICU_d1d2d3_full)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.1190  -3.1190   0.2976   1.8810   7.8810
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  17.1190     0.4179  40.965  < 2e-16 ***
## diaryTP2      1.5833     0.5910   2.679  0.00787 **
## diaryTP3      1.0238     0.5910   1.732  0.08445 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.83 on 249 degrees of freedom
## Multiple R-squared:  0.0288, Adjusted R-squared:  0.021
## F-statistic: 3.692 on 2 and 249 DF,  p-value: 0.0263
```

```
##                  2.5 %    97.5 %
## (Intercept) 16.2959921 17.942103
## diaryTP2     0.4193571  2.747310
## diaryTP3    -0.1401667  2.187786
```

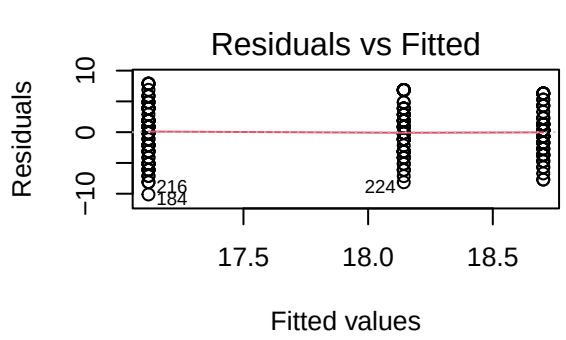
```
##                  0.25 %   99.75 %
## (Intercept) 15.93546320 18.302632
## diaryTP2    -0.09050781  3.257174
## diaryTP3    -0.65003162  2.697651
```

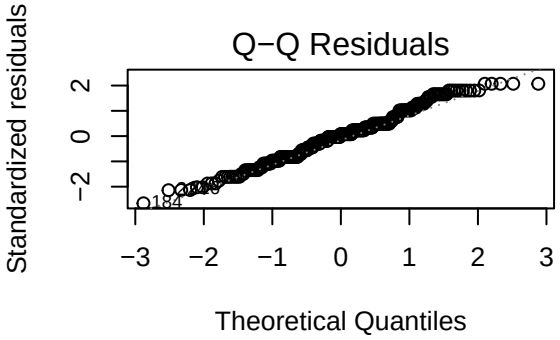
Standardised parameters:

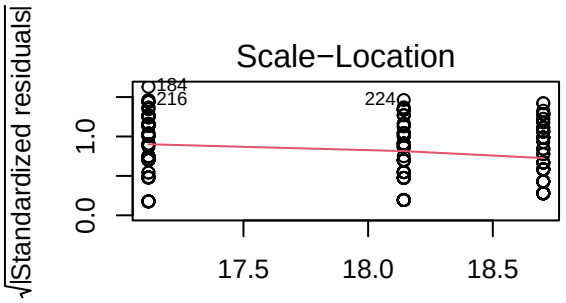
```
## Parameter   | Coefficient |   SE |         95% CI | t(249) |     p
## ------------------------------------------------------------------
## (Intercept) |       -0.22 | 0.11 | [-0.44, -0.01] |  -2.08 | 0.039
## diaryTP [2] |        0.41 | 0.15 | [ 0.11,  0.71] |   2.68 | 0.008
## diaryTP [3] |        0.26 | 0.15 | [-0.04,  0.57] |   1.73 | 0.084
```

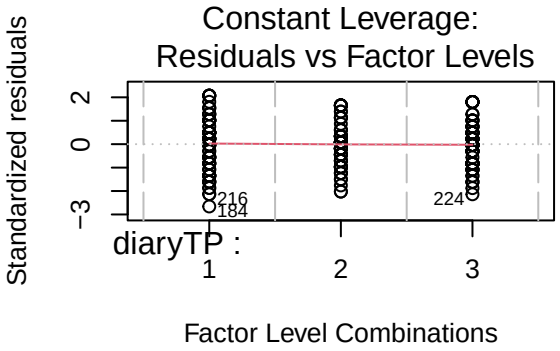
```
##
## Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
##   using a Wald t-distribution approximation.
```

```
## # Standardization method: refit
##
## Parameter   | Std. Coef. |      99.5% CI
## ----------------------------------------
## (Intercept) |      -0.22 | [-0.53, 0.08]
## diaryTP [2] |       0.41 | [-0.02, 0.84]
## diaryTP [3] |       0.26 | [-0.17, 0.70]
```


Residuals vs Fitted
Residuals
Fitted values
216
184
224


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
184


Scale−Location
√|Standardized residuals|
Fitted values
184
216
224


Constant Leverage:
Residuals vs Factor Levels
Standardized residuals
diaryTP :
Factor Level Combinations
216
184
224

6.3) (exploratory) Do teacher ratings of help differ from T1-3

```
##
## Call:
## lm(formula = T_TSRI_Help_Total ~ diaryTP, data = TSRI_ICU_d1d2d3_full)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.0595 -3.5774 -0.2262  3.7738 11.7738
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  13.2262     0.5911  22.376   <2e-16 ***
## diaryTP2      0.1905     0.8359   0.228     0.82
## diaryTP3      0.8333     0.8359   0.997     0.32
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.417 on 249 degrees of freedom
## Multiple R-squared:  0.004364,   Adjusted R-squared:  -0.003633
## F-statistic: 0.5457 on 2 and 249 DF,  p-value: 0.5801
```

```
##                  2.5 %    97.5 %
## (Intercept) 12.0620142 14.390367
## diaryTP2    -1.4559178  1.836870
## diaryTP3    -0.8130606  2.479727
```

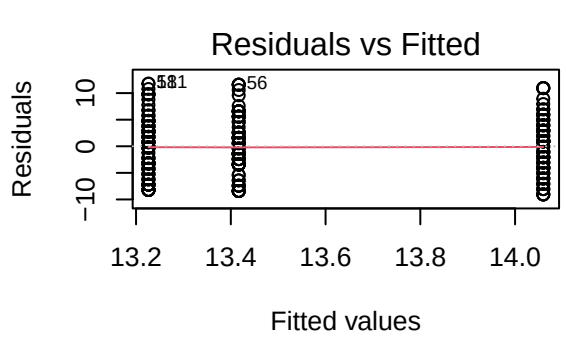
```
##             -4900 % 5000 %
## (Intercept)     NaN    NaN
## diaryTP2        NaN    NaN
## diaryTP3        NaN    NaN
```

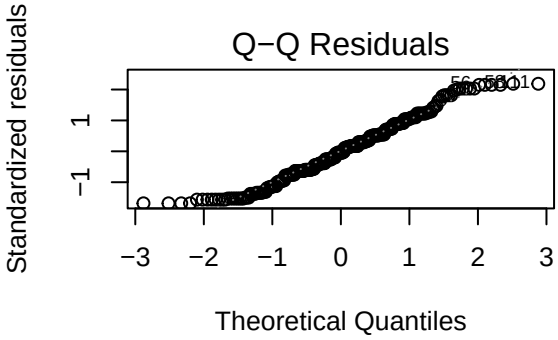
```
## Parameter   | Coefficient |   SE |        95% CI | t(249) |     p
## ------------------------------------------------------------------
## (Intercept) |       -0.06 | 0.11 | [-0.28, 0.15] |  -0.58 | 0.564
## diaryTP [2] |        0.04 | 0.15 | [-0.27, 0.34] |   0.23 | 0.820
## diaryTP [3] |        0.15 | 0.15 | [-0.15, 0.46] |   1.00 | 0.320
```

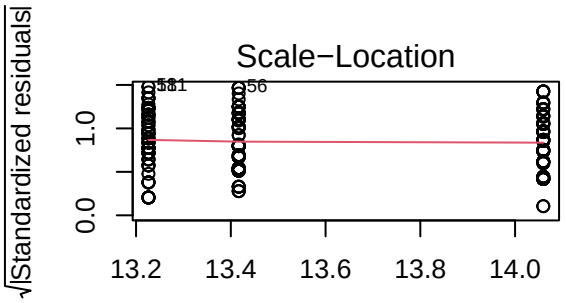
```
##
## Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
##   using a Wald t-distribution approximation.
```

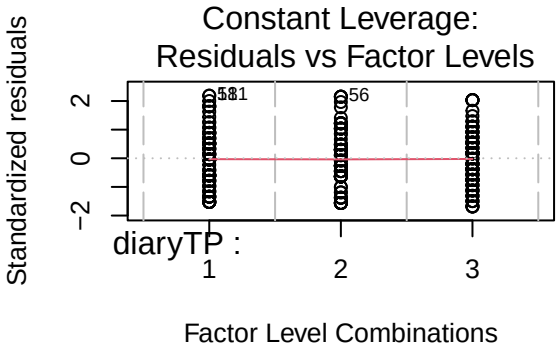
```
## Parameter   | Coefficient |   SE |      99.5% CI | t(249) |     p
## --------------------------------------------------------------------
## (Intercept) |       -0.06 | 0.11 | [-0.37, 0.25] |  -0.58 | 0.564
## diaryTP [2] |        0.04 | 0.15 | [-0.40, 0.47] |   0.23 | 0.820
## diaryTP [3] |        0.15 | 0.15 | [-0.28, 0.59] |   1.00 | 0.320
```

```
##
## Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
##   using a Wald t-distribution approximation.
```

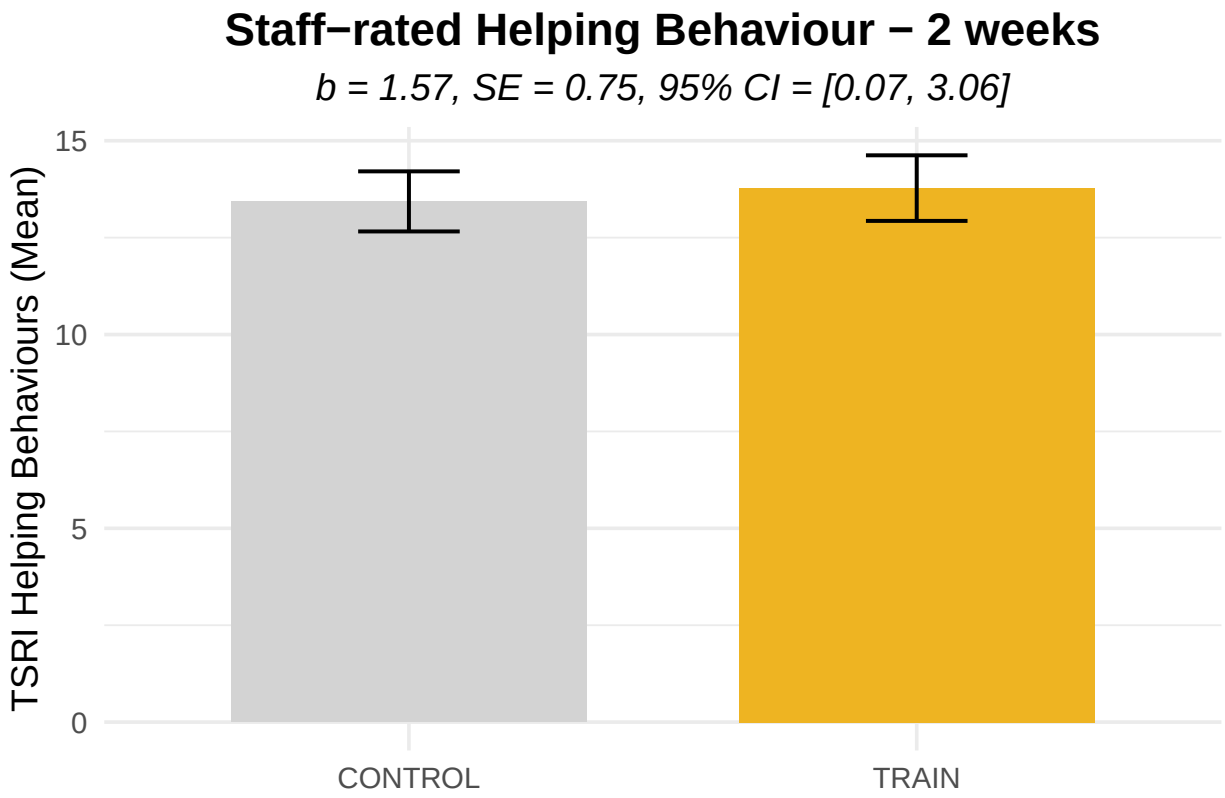

Residuals vs Fitted
Residuals
Fitted values
13.2
13.4
13.6
13.8
14.0
−10
0
10
56

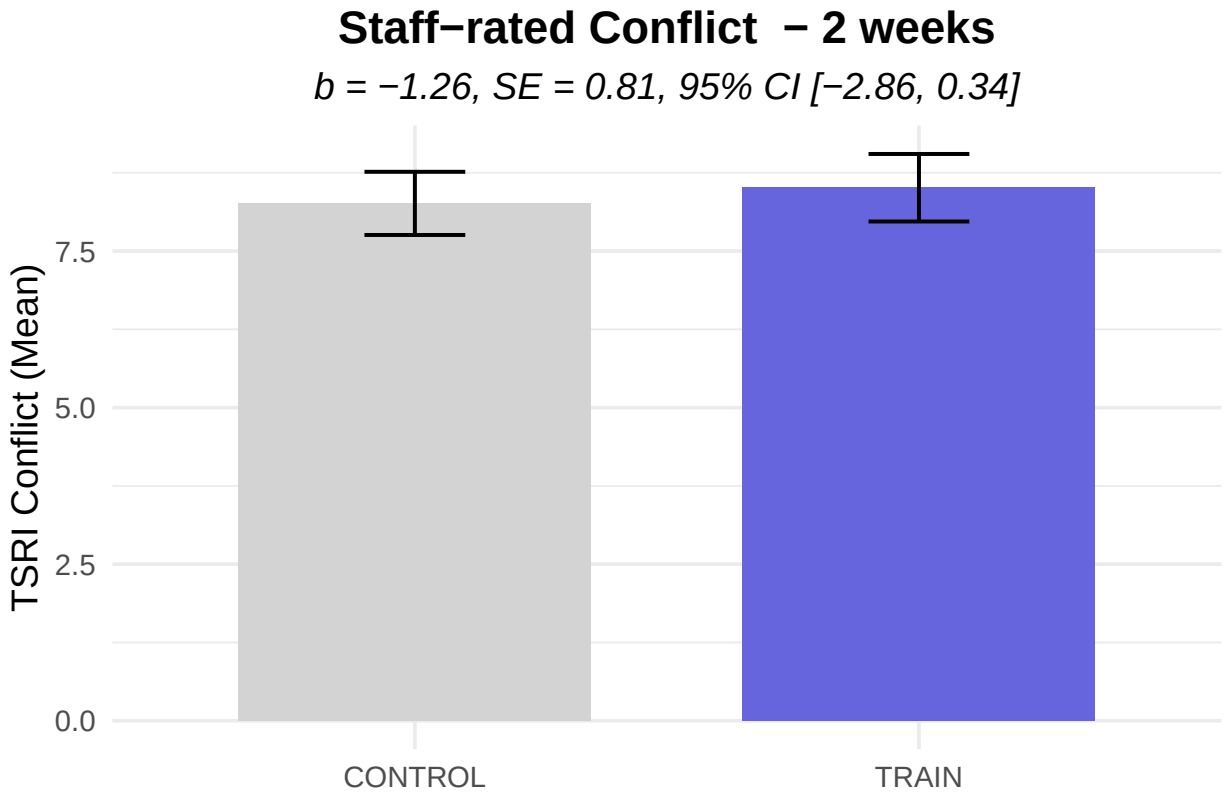

Q−Q Residuals
Standardized residuals
Theoretical Quantiles
−3
−2
−1
0
1
2
3


Scale−Location
√|Standardized residuals|
Fitted values
13.2
13.4
13.6
13.8
14.0
0.0
1.0
56


Constant Leverage:
Residuals vs Factor Levels
Standardized residuals
diaryTP :
Factor Level Combinations
1
2
3
−2
0
2
56

extra: TSRI Bar plots (for emodev lab presentation)






Staff−rated Satisfaction − 2 weeks
b = −1.89, SE = 1.09, 95% CI = [−4.06, 0.27]
TSRI Satisfaction (Mean)
20
15
10
5
0
CONTROL
TRAIN